

Jacobi Method

The Jacobi method is an iterative algorithm used to solve a system of linear equations. The general form of a system of linear equations is:

$$A\mathbf{x} = \mathbf{b}$$

where A is the coefficient matrix, $\mathbf{x}$ is the vector of unknowns, and $\mathbf{b}$ is the right-hand side vector.

For a given system of equations:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = b_n \end{cases}$$

the Jacobi method updates the value of each unknown x_i based on the previous values of all the other unknowns. The formula for updating x_i is:

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j \neq i} a_{ij}x_j^{(k)} \right)$$

where k denotes the iteration step.

Problem

Given the system of equations, find the solution approximation after 2 iterations of the Jacobi method:

$$\begin{cases} 60x_1 - 30x_2 - 20x_3 = -400 \\ -30x_1 + 180x_2 - 60x_3 = 400 \\ -20x_1 - 60x_2 + 150x_3 = 300 \end{cases}$$

we can rewrite the equations in the form suitable for the Jacobi method. Starting with an initial approximation $(x_1^{(0)}, x_2^{(0)}, x_3^{(0)}) = (0, 0, 0)$:

1. Initial Values:

$$x_1^{(0)} = 0, \quad x_2^{(0)} = 0, \quad x_3^{(0)} = 0$$

2. First Equation:

$$x_1^{(k+1)} = \frac{1}{60} \left(-400 + 30x_2^{(k)} + 20x_3^{(k)} \right)$$

3. Second Equation:

$$x_2^{(k+1)} = \frac{1}{180} \left(400 + 30x_1^{(k)} + 60x_3^{(k)} \right)$$

4. Third Equation:

$$x_3^{(k+1)} = \frac{1}{150} \left(300 + 20x_1^{(k)} + 60x_2^{(k)} \right)$$

5. First Iteration ($k = 0$):

$$x_1^{(1)} = \frac{1}{60} (-400 + 30 \cdot 0 + 20 \cdot 0) = \frac{-400}{60} = -\frac{20}{3} \approx -6.6667$$

$$x_2^{(1)} = \frac{1}{180} (400 + 30 \cdot 0 + 60 \cdot 0) = \frac{400}{180} = \frac{20}{9} \approx 2.2222$$

$$x_3^{(1)} = \frac{1}{150} (300 + 20 \cdot 0 + 60 \cdot 0) = \frac{300}{150} = 2$$

So after the first iteration, we have:

$$x_1^{(1)} \approx -6.6667, \quad x_2^{(1)} \approx 2.2222, \quad x_3^{(1)} = 2$$

6. Second Iteration ($k = 1$):

$$x_1^{(2)} = \frac{1}{60} (-400 + 30 \cdot 2.2222 + 20 \cdot 2) = \frac{1}{60} (-400 + 66.666 + 40) = \frac{-293.334}{60} \approx -4.889$$

$$x_2^{(2)} = \frac{1}{180} (400 + 30 \cdot -6.6667 + 60 \cdot 2) = \frac{1}{180} (400 - 200 + 120) = \frac{320}{180} \approx 1.778$$

$$x_3^{(2)} = \frac{1}{150} (300 + 20 \cdot -6.6667 + 60 \cdot 2.2222) = \frac{1}{150} (300 - 133.334 + 133.332) = \frac{299.998}{150} \approx 2$$

So after the second iteration, we have:

$$x_1^{(2)} \approx -4.889, \quad x_2^{(2)} \approx 1.778, \quad x_3^{(2)} \approx 2$$

Thus, the solution approximation after 2 iterations of the Jacobi method is:

$$(x_1, x_2, x_3) \approx (-4.889, 1.778, 2)$$